Operation Contracts:

UC-01: Book Appointment

Contract CO-01-1: bookAppointment
Operation: bookAppointment(dateTime)
Cross References: UC-01 Book Appointment

Preconditions:

- The patient is registered in the system and authenticated or has been selected by administrative staff
- A physician/healthcare professional and service type have been selected
- A valid referral exists if the service requires one.
- The selected professional is available at the requested date and time.

Postconditions:

- An Appointment instance *appointment* was created. (instance creation)
- *appointment.appointmentId* was assigned a unique identifier. (attribute modification)
- *appointment.dateTime* was set to *dateTime*. (attribute modification)
- *appointment.status* was set to confirmed. (attribute modification)
- *appointment.serviceType* was set to the selected service type. (attribute modification)
- *appointment* was associated with the selected *Patient*. (association formed)
- *appointment* was associated with the selected *HealthcareProfessional*. (association formed)
- *appointment* was associated with the *ClinicLocation* where the appointment takes place. (association formed)
- If a referral was required, *appointment* was associated with its *Referral*. (association formed)

UC-07: Register Payment

Contract CO-07-1: registerPayment
Operation: registerPayment(invoiceId, amount, paymentDate, paymentMethod)
CrossReference: UC-07 Register Payment

Preconditions:

- The patient is registered in the system
- The invoice exists
- The payment details are valid

Postconditions:

- A Payment instance *payment* was created. (instance creation)
- *payment.paymentId* was assigned a unique identifier. (attribute modification)
- *payment.amount* was set to *amount*. (attribute modification)
- *payment.paymentDate* was set to *paymentDate*. (attribute modification)
- *payment.paymentMethod* was set to *paymentMethod*. (attribute modification)
- *payment* was associated with the selected *Invoice*. (association formed)
- The *Invoice* status was updated according to the payment and remaining balance. (attribute modification)

UC-08: Issue Prescription

Contract CO-08-1: IssuePrescription
Operation: issuePrescription()
Cross References: UC-08 Issue Prescription.

Preconditions:

- The patient's clinical record exists.
- The selected consultation exists.
- A prescription draft exists.
- At least one valid medication has been added.
- All required prescription details are complete and valid.

Postconditions:

- A Prescription instance *prescription* was created. (instance creation)
- *perscription.perscriptionId* was assigned a unique identifier. (attribute modification)
- *prescription.issueDate* was set to the current date. (attribute modification)
- *prescription* was associated with the selected *Consultation*. (association formed)
- For each medication in the draft, a PrescriptionItem instance *prescription_item* was created. (instance creation)
- *prescription_item.dosage* was set to *dosage*. (attribute modification)
- *prescription_item.frequency* was set to *frequency*. (attribute modification)
- *prescription_item.duration* was set to the *duration*. (attribute modification)
- Each *prescription_item* was associated with the *prescription*. (association formed)
- Each *prescription_item* was associated with the selected *Medication*, based on *medicationId*. (association formed)